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# Detail-aware semantic segmentation network for brain tumor MRI images combining multi-frequency directional filtering and lifting wavelets

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## Highlights

- Wavelet-based Unet for brain tumor segmentation developed using frequency information.
- Develop a novel downsampling method centered on lifting wavelets.
- Designing a dual-channel directional filtering module to acquire feature orientation information.
- Our model outperforms other advanced models in segmenting two brain tumor datasets.

## Abstract

Brain tumors segmentation in Magnetic Resonance Imaging (MRI) images poses significant challenges owing to the uncertain location and size of the tumors, the difficulty in describing their boundaries, and the fuzzy demarcation of diseased tissues. Although U-Net and its recent variants have emerged as leading models for semantic segmentation in medical imaging, they still face structural limitations. These limitations cause the erosion of detail information during downsampling and poor performance in segmenting small lesions when handling targets of varying sizes, indicating a lack of detail handling capability. To counteract these issues, we designed a segmentation model that enhances detail features using frequency information. To reduce the loss of feature information during downsampling, we developed a downsampling module based on lifting wavelets. By lifting wavelets to group and integrate features according to frequency from high to low, we reduce feature resolution while enhancing information transmission and minimizing feature information loss. In our designed multi-frequency directional filtering edge feature extraction module, we extract low-frequency and high-frequency features and construct a dual-channel multi-directional filtering combination. This combination extracts directional information from low-frequency and high-frequency features separately, increasing the multi-angle directional information of the features and enriching the detailed information such as direction and position within the features. On the BraTS2018, BraTS2020, and BraTS2024 brain tumor datasets, our model demonstrated optimal results compared to 14 other advanced models. The average Dice Similarity Coefficients are 78.48 %, 79.80 %, and 74.35 %, while the 95th percentile Hausdorff Distances are 5.75, 6.60, and 7.72. Our code link is https://github.com/Eric-H8/BraTS_Seg_Model.



## Introduction

Accurate segmentation of brain tumors is crucial for aspects such as treatment planning, diagnosis, and tracking, as the location of brain tumors can potentially impact surrounding functional areas [1], [2], [3]. Brain tumor segmentation is more complex compared to other organ segmentation tasks (such as the heart, liver, etc.). This is attributable to the stochastic nature of brain tumors with respect to their shape, size, and location within the brain tissue. Additionally, there is often no clear boundary between brain tumors and normal brain tissue, making the segmentation task even more challenging.



Amid the rapid advancements in deep learning technology, convolutional neural networks (CNN) have demonstrated the capability to automatically extract key features networks (CNN) have demonstrated the capability to automatically extract key features and obtain semantic information, eliminating the need for manual feature annotation and predefined template design. However, CNNs have issues such as complex network structures, the possibility of getting stuck in local minimum during optimization, and the tendency to overfit when training data is insufficient, all of which can lead to decreased generalization ability on new data [4]. To address these issues, researchers have developed various new structural models. Among them, the U-Net network [5], with its symmetrical structure comprised of an encoder and a decoder, has shown superior performance in medical image segmentation tasks. However, due to inherent structural limitations, U-Net's receptive field and network depth are insufficient, making it difficult to segment complex targets. To address this issue, researchers designed ResUNet [6], which extends the receptive field by increasing network depth. Building on this, models like UNet++ [7]and ADS_Unet [8] were developed, which adopt dense skip connections to acquire multi-scale information while enhancing the sensory field and depth of the network. Additionally, models such as DCSAU-Net [9] and MhorUNet[10] use large convolutional kernels to increase the model's receptive range. However, these methods still do not fully solve the problems of insufficient receptive field and network depth [11]. The downsampling operations used in these networks, such as strided convolutions or max pooling, can aggregate local features and expand the receptive field while minimizing computational overhead. However, for semantic segmentation tasks, these operations may lead to the diminishment of significant spatial information such as boundaries and textures, which is detrimental to pixel-level predictions.

With the outstanding performance of ViT[12] in the branch of computer vision, researchers have introduced it into the U-Net network structure, leading to the development of models such as RTUnet[13], UCTNet[14], and UCFilTransNet[15]. These networks leverage the excellent long-range modeling capabilities of ViT to address the issues of limited receptive field and depth in models. However, due to the core Multi-Head Self-Attention (MHSA) mechanism in ViT, its computational complexity and limited receptive field hinder its ability to capture local information. This results in hybrid Transformer methods being computationally complex and having redundant dependencies [16]. Therefore, some researchers have considered incorporating frequency domain information by analyzing the dispersion of diverse frequency elements across the features. Due to the characteristics of brain tumors in MRI images, such as random positions, irregular edges, and the fuzzy boundaries of various tissues, frequency domain features can provide rich frequency information [17]. Additionally, the frequency distribution of targets and backgrounds in the texture information of images typically differs. By integrating frequency features with other features in the model, it is possible to effectively and intricately describe the texture information. This can help the model better distinguish between different objects, thereby enhancing the model's expressiveness and adaptability. This approach helps the network understand critical information such as image texture and boundaries, both globally and locally. Integrating frequency spectrum information into the model can provide more exhaustive feature information, thereby elevating segmentation exactness.

To more accurately segment brain tumors in MRI images, we have constructed a novel U-shaped network that integrates frequency feature information with the U-shaped architecture. By combining frequency domain information with spatial information, the network's capability to capture detailed information is enhanced, thereby improving the model's segmentation performance. We employed a multi-residual structure in the encoder and decoder of the network to enhance network depth and increase the receptive field. For the downsampling part, we designed a downsampling module based on lifting wavelets, which groups and integrates features from a frequency perspective using lifting wavelets. This approach strengthens information transmission and reduces feature information loss. In the skip connections, we designed a multi-frequency directional filtering edge feature extraction module. This module uses a combination of multi-directional filters to achieve directional feature extraction, extracting high-frequency and low-frequency information separately. This enhances directional information and strengthens boundary and other detailed information, thereby improving the model's capability to capture delicate details. We summarized our main contributions as follows:

(1) To address the random characteristics of brain tumors in reference to shape, size, and boundary, we have designed a novel frequency feature semantic segmentation model. This model combines the U-Net architecture with various filters, including the lifting wavelet transform, leveraging both frequency domain information and spatial information. This approach effectively captures the intricate boundaries and complex shapes of tumors, thereby enhancing segmentation performance.

(2) We designed a novel feature downsampling module that employs lifting wavelet for the downsampling operation. By computing on the features themselves, this module further extracts and separates the low-frequency and high-frequency information, constructing different frequency feature representations. This ensures that the module can maintains as much information as possible during the downsampling process, thereby enhancing the learning performance of the model.。

(3) We designed a multi-frequency directional filtering edge feature extraction module (MFDF). By creating new directional gradient feature extraction

operators and using dual-channel filtering to combine operators in different directions, we constructed multi-directional filters to separately extract low-frequency and high-frequency feature directional information. This approach not only enhances the feature directional information but also increases the contrast between high-frequency and low-frequency information.

(4) The model we designed achieved excellent segmentation performance, outperforming other advanced semantic segmentation models, on three publicly available brain tumor datasets, BraTS2018, BraTS2020 and BraTS2024, using the Dice Similarity Coefficient (DSC) and 95th Percentile Hausdorff Distance (HD95) evaluation metrics.



## Section snippets

### Application of U-net and its variants in semantic segmentation

In recent years, the U-net network has accomplished remarkable success in the domain of medical image segmentation. However, due to inherent structural limitations, its segmentation capabilities are restricted. Consequently, many researchers have been dedicated to optimizing and improving the U-net structure. RU-Net [18] and R2U-Net [18] combine U-Net with residual networks and recursive convolutional neural networks (RCNN), achieving multi-layer feature accumulation and enhancing the network's …

### Overview

The architecture of our designed model is depicted in Fig. 2. The overall architecture adopts a U-Net structure, with the encoder and decoder incorporating multiple residual network structures to enhance the sensory field and capture spatial features of the image. To reduce feature information loss during downsampling, an uplifting wavelet downsampling module is used. The features from the encoder are passed through skip connections into a multi-frequency directional filtering edge feature …

### Datasets

To validate the efficacy of our designed model, we selected three public brain tumor datasets: BraTS2018[35], BraTS2020[36], and BraTS2024[37], for training and validation. These datasets include brain MRI images from 285, 369, and 1350 patients, respectively, with each image comprising four modalities: T1, T1ce, T2, and FLAIR. All three datasets include annotations for necrotic tumor core (NCR), peritumoral edema/invasive tissue (ED), and enhanced tumor (ET), with the BraTS2024 dataset …

### Conclusion

We address the challenge of the random and uncertain sizes and shape boundaries of brain tumors by designing a semantic segmentation model that combines a U-shaped network with frequency features. In our model, we integrate wavelet transform into the down-sampling module to extract and process high-frequency and low-frequency information during down-sampling. This approach maximizes the retention of effective feature information and minimizes information loss, thereby enhancing the learning …

### CRediT authorship contribution statement

**Hui Zhao:** Resources, Data curation. **Qiaohui Lu:** Data curation, Resources. **Zibo Li:** Writing – review & editing. **Hongjian Yu:** Supervision. **Zhijiang Du:** Supervision. **Xin Hua:** Methodology, Writing – original draft, Conceptualization, Software. …

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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